MTH-202 Quiz File

By Knowledge for us

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1. In the proof by contradiction, we lead the assumption to___.

   Answer: absurdity

2. The predicate describing the initial state is called____.

   Answer: pre-condition of the algorithm

3. An integer n is prime number, if and only if for all positive integers r and s, if n= rs then____.

   Answer: r = 1 or s = 1.

4. If ‘f’ is a perfect square, then which of the following is the correct representation of ‘r’?

   Answer: r = k^2 for some integer k

5. Which of the following is not a rational number?

   Answer: $\sqrt{7}$

6. To prove by mathematical induction for 1+2+3+...+n= (n) (n+1)/2 for all positive integers, the basis step is____.

   Answer: 1 = (1) (1+1)/2

7. GCD of (8, 12) will be____.

   Answer: 4

8. The statement $4^n > 3^n + 4$ is true when____.

   Answer: $n \geq 2$

9. If n is any positive integer then $2^n \geq 2$ (n+1) is true for all_____.

Answer: $n \geq 3$

10. The sum of $(6-7\sqrt{2})$ and $(6+ 7\sqrt{2})$ is____.

Answer: 12

11. The sum of two integers is not an even integer when____.

Answer:

12. If n is an add integer then '3n+2' is____.

Answer: odd

13. The sum of any rational number and any irrational number is irrational.

Answer: true

14. A proof by___is based on the logical equivalence between a statement and its contrapositive.

Answer: contrapositive

15. For every prime number n, n+2 is prime.
Which of the following prime number disproves the above statement.

Answer: n=7

16. In basis property the first iteration of the loop is solved for?

Answer: I (0).

17. For all integers n>0
1+2+2^2+___+2^n=___.

Answer: 2^ (n-1)

18. In division algorithm 'r' stands for____.

Answer: remainder.

19. If n is any positive integer then the sequence 3+6+9+-----+3n =_____.

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Answer: $\frac{3n(n+1)}{2}$

20. For all positive real numbers a and b, if a<b, then____.

Answer: a^2 < b^2

21. The last step of describing algorithms formally is?

Answer: An end statement

22. If 'n' is an odd integer then n^3+n is___

Answer: even

23. Proof by____uses the equivalence p→q = -q→ p.

Answer: Contraposition

24. The product of an even and odd integer is____.

Answer: even

25. In the Direct Proof, we show that if the statement p is true, then the statement q is___.

Answer: True

26. (-1)^n = 1 for n = ___.
(^ stands for power)

Answer: 4

27. Which of the following is true for n = 1?

Answer: 2^ (2n) – 1 by 3

28. Proof by____uses the equivalence p→q =(p∧ −q)→ c.

Answer: Contradiction

29. Proof by contradiction is a/an____method of proof.

Answer: Indirect.

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30. An integer ‘n’ is____if it can be represented as a multiple of 2.

Answer: even.

31. The contrapositive of ‘5n+2 is odd, then n is odd’ is___.

Answer; “if n is even then 5n+2 is even”

32. Which of the following is not a method to prove the mathematical statements?

Answer: Permutation.

33. The sum of two integers is not an even integer when____.

Answer: one is even and one is odd.

34. The sum of two odd integer’s is____.

Answer: even

35. The predicate which describes the initial state is called____of the algorithm.

Answer: Pre-condition

36. Algorithm is a more general term in that term____refers to a particular programming language.

Answer: Program

37. In division algorithm the pre-condition is?

Answer: A is a nonnegative integer and d is a positive integer.

38. Basis and inductive steps are part of proof by____.

Answer: Mathematical induction

39. If A and B are disjoint sets then P (A B) =……

Answer: P (A)

40. If B is a subset of A then P(A/B)= 1

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Answer: True

41. $P(A^C/B)=1 - P(A/B)$

Answer: **True**

42. If A and B be events with P (A) =1/3. P (B) =1/4 and P (A intersection b)=1/6, then P(A U B) = _____.

5/12

43. P (A/B) = P (A n b)/P (B), where _____.

P (B) > 0

44. Total degree of a graph G is _____.

2(Number of edges of G)

45. Let $E(X^2)$ =6, u^2=2, then standard deviation is _____.

2

46. If A and B be events with P (A) =1/3. P(B)=1/4 and P(A intersection b)=1/6, then P(A/B) = _____

2/3

47. A closed walk is the walk that starts and ends at the same vertex

True

48. Total degree of G is an odd number

False

49. If X and Y are random variable, then E(aX) is equal to

aE (X)

50. A simple graph is a graph that does not have any loop.

True

51. Conditional probability of A given B is defined as _____.

P (A n B) / P (B)

52. If X and Y are independent random variables and a and b are constant, then Var(aX+bY) is equal to

A^2 Var(X) + b^2Var(Y)

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53. A random variable is also called a

Chance variable

54. If the random variable X denotes the number of heads when three distinct coins are tossed, then X assumes the value

0,1,2,3

55. If X and Y are random variables, then E (X-Y) is equal to

E(X) - E(Y)

56. A random variable is also called a Stochastic variable

True

57. Which of the followings is the probability of getting a 6 when a dice is tossed?

1/6

58. A random variable is also called a

Chance variable or stochastic variable

59. If E(5)=5, then find arithmetic mean will be

5

60. Conditional probability of B given A is defined as _____.

P (A n B) / P (A)

61. If A is a subset of B then P(A I B) = P(A)/P(B)

True

62. A graph is regular if every vertex has the _____ degree

Same

63. If the standard deviation of the data set is 4 then the value of the variance is

16

64. What is the probability of the number of one head when two fair coins are tossed?

2/4

65. A simple graph is a graph that does not have any loop

True

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66. If A and B are disjoint sets then P (A/B) = _____?

P (A)

67. A complete graph on n vertices is a simple graph in which every is connected to every other vertex.

True

68. There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

6

69. Which of the followings is the product set A*B*C? where A= {a}, B={b}, and C={c,d}

{(a, b, c), (a, b, d)}

70. In how many ways can 6 people be seated on 6 available seats?

720

71. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

16

72. Compute C(8 , 3)

56

73. Let X= {1, 2, 3}, then 2-combinations of the 3 elements of the set X are _____?

{1, 2}, {1, 3} and {2, 3}

74. P (A n B) = P (A). P(A/B)

False

75. If B is a subset of A then P(A/B) = 1

True

76. An expected value of a random variable is equal to its mean

True

77. If X and Y are independent random variables, then E(XY) is equal to

E(x) E(y)

78. A random variable is also called stochastic variable

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True

79. If A and B are two independent events then P(A n b) =

P (A) P (B)

80. If X is a discrete random variable and f(x) is the probability of X, then the expected value of this random variable is equal to;

$\sum xf(x)$

81. The expected value E(X) is obtained by multiplying each value of X by its probability and taking the sum.

True

82. P (A, A) = _____.

1

83. Which of the followings is the factorial form of 5 . 4?

5!/3!

84. The set of prime numbers is _____.

Infinite set

85. Let R be a relation on a set A. If R is symmetric then its compliment is _____.

Irreflexive

86. There are 4 bus lines between X and Y, and 5 bus lines between Y and Z, the number of ways a person can travel from X to Z?

20

87. To prove by mathematical induction inductive step is _____.

If P (K) is true then P(K + 1) is true for all integers $K \geq 1$

88. Algorithm is a more general term in that the term ____ refers to a particular programming language.

Program

89. The sum of rational and irrational number _____.

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Is always an irrational number

90. Basis and Inductive steps are part of proof by __.

Mathematical induction

91. Reduction and absurdum is another name of ___.

Proof by contradiction

92. If one event can occur in n_1 ways a second event can occur in n_2 ways, a third event can occur in n_2 ways, how ways all of the events can occur in the order

$N_1+n_2+n_3$

93. A non-zero integer d divides an integer n if and only if there exists an integer k such that_____

N=dk

94. The sequence 1+2+3+4 --------- +n is equal to _____.

N (n+1)/2

95. The sum of two integers is not an even integer when _____.

One is even and one is odd

96. To prove by mathematical induction for 1+2+3+…+n= (n) (n+1)/2 for all positive integers, the basis step is _____.

1=(1)(1+1)/2

97. To prove by mathematical induction for 1+5+9+…. + (4n-3) =n (2n-1) for all positive integers, the inductive step is true for n=k is _____.

1+5+9+……. + (4k-3) = k(2k-1)

98. Proof by _____ uses the equivalence p -> q = ~q -> ~p

Contraposition

99. The predicate which describes the initial state is called the _____ of the algorithm

Pre-condition

100. The last step of describing algorithms formally is?

An end statement

101. Which of the following is not a method to prove the mathematical statements?

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Permutation

102. GCD of (330.156) is?

6

103. Which of the following is not an irrational number?

Square root (4)

104. The sum of a rational and irrational number ___.

Is always an irrational number

105. If n is an odd integer then n^3+n is _____.

Even

106. The ___ of a predicate variable is the set of all values that may be submitted in place of the variable

Domain

107. For all positive real numbers a and b if a < b, then _____.

A^2 < b^2

108. For every prime number n, n+2 is prime. Which of the following prime number disproves the above statement.

N=7

109. The set of prime number is _____.

Infinite set

110. An integer n is _ if it can be represented as a multiple of 2

Even

111. In inductive property the first iteration of the loop is solved for?

I (K)

112. Which of the following is not a method to prove the mathematical statements?

Permutation

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113. In how many ways a student can choose a course from 2 science courses 3 literature courses and 5 art courses

10

114. In the Direct Proof, we show that if the statement p is true, then the statement q is _____

True

115. Which of the following is true for n = 2?

X^n – y^n is divisible by (x – y) ^ 2

116. The sum of any rational number and any irrational number is irrational.

True

117. The set of prime numbers is _____.

Infinite set

118. The sum of two odd integers is _____--.

Even

119. How many multiples of 5 are there from 10 to 75?

14

120. A bit string is a sequence of 0's and 1's.

True

121. The method of loop invariants is used to prove _____ of a loop with respect to certain pre and post-conditions.

Correctness

122. _____ is used to determine number of ordered or unordered arrangement of objects

Counting

123. While proofing by contradiction the equivalence _____ is used.

P →q ≡ (p∧~q) →c

124. While proofing by contraposition the equivalence _____ is used.

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P →q≡ ~q→~p

125. Proof by contradiction is a/an ___ method of proof.

Indirect

126. Proof by contraposition is a/an ___ method of proof.

Indirect

127. If n is divisible by 5, then n^2 is divisible by 25 is the contraposition of _____.

N^2 is not divisible by 25, then n is not divisible by 5

128. The quantity a – d. q equals the?

Remainder

129. In inductive step of proof by mathematical induction, the assumption is made for ____.

N=k+1

130. How many 4-bit string contain at least one '1'

16

131. For all integers n>0
1+2+2^2+…..+2^n=_____

2^(n+1) – 1

132. The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit L letters are a and b and numbers are 3 and 5 what is the largest number of chairs that can be labeled differently?

15

133. To prove by mathematical induction for 1+5+9+….+ (4n-3)=n(2n-1) for all positive integers, the basis step is _____.

1=1(2(1)-1)

134. The predicate describing the final state is called _____.

Post condition of the algorithm

135. The predicate describing the initial state is called _____.

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Pre-condition of the algorithm

136. If n is any positive integer then $2^n \geq 2(n+1)$ is true for all _____.

N ≥ 3

137. The sum of (6 – 7 under root (2)) and (6 + 7 under root(2)) is

12

138. GCD of (8, 12) will be _____.

4

139. In division algorithm ‘r’ stands for ____

Remainder

140. If the sum of two integers is even the difference would be _.

Even

141. Which of the following is true for n=1?

2^ (2n) – 1 by 3

142. To prove by mathematical induction for 1+3+5+….+(2n-1)=n^2 for all positive integers, the basis step is _____.

LHS=p(1)=2x1=1, RHS = 1^2=1

143. The loop is correct if the four properties are true, one of the property is?

Eventual feisty of guard

144. The statement $n^2 > n + 3$ is true when _____.

N ≥ 3

145. If ‘n’ is an odd integer then n^3+n is _____.

Even

146. (-1)^n=1 for n = (^standard for power)

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4

147. A boy can choose from 5 train services and 3 bus service to go to his hometown, How many total options he can choose from?

8

148. The method of loop invariants is based on the people of _____.

Division algorithm

149. The statement $3^n < n!$ is true when _____.

n>6

150. In a city, the bus route numbers consists of a natural number less than 20, followed by one of the letters A, B, C, D, E and F. How many different bus routes are possible

114

151. The word algorithm refers to a step-by-step method for performing some action.

True

152. If n is an odd integer then 3n+2 is _____.

Odd

153. To prove by mathematical induction basis step is _____.

The proposition P (1) is true

154. There are 5 Chinese books and 6 English books, a student wants to select one optional book for both subject, total number of choices will be

30

155. In division algorithm the pre-condition is?

A is a nonnegative integer and d is a positive integer

156. If there are 5 different optional courses in English and 3 different optional in Maths. Choices for a student who to take one optional.

8

157. Number of combination of 3-bit string that contain exactly there ‘0’ is _____.

8

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158. The sequence $1^3+2^3+3^3+\ldots\ldots+n^3$ = _____.

N (n+1) (2n+1)/6

159. If 'r' is a perfect square, then which of the following is the correct representation of 'r'?

R=k^2 for some integer k

160. Which of the following is the representation of an odd integer 'n'?

N=2k+1 for some integer

161. If n is any positive integer then the sequence 3+6+9+……..+3n=_____.

3n (n+1)/2

162. A predicate becomes _____ when its variables are given specific values.

Sentence

163. The greatest common divisor (gcd) of two integers a and b is the largest integer that divides both a and b. it is called _____.

EUCLIDEAN ALGORITHM

164. A predicate is a sentence that contains a finite number of_____.

Variables

165. A proof by _____ is based on the fact that either a statement is true or it is false but not both.

Contradiction

166. The _____ of a predicate variable is the set of all values that may be substituted in place of the variable

Domain

167. Which of the following statement is true according to the division algorithm?

17=15 x 3 + 2

168. The division by zero is allowed in mathematics

True

169. A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

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170. For integers a,b,c if a divides b and a divides c, then a divides (a + b)

True

171. Suppose there are different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

13

172. The sequence 1+2+3+4……+ n is equal to _____.

N (n+1)/2

173. The product of two odd integers is _____.

Odd

174. A student has 3 optional courses in mathematics and 5 optional courses in physics, how many total choices for him to take one course?

8

175. If the square of an integer is even, then that integer is _____.

Even

176. The first step of describing algorithm formally is?

The name of the algorithm, together with a list of input and output variables

177. Set of prime numbers is finite.

false

178. The sum of two irrational numbers in general _____.

Is always a rational number

179. Which of the following is the representation of an odd integer ‘n’?

N=2k+1 for some integer ‘k’

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180. In how many ways can 6 people be seated on 6 available seats?

12

181. The product of an even and odd integer is _____.

Even

182. In the direct proof, we show that if the statement p is true then the statement q is _____.

True

183. _____ is the mathematics of counting and arranging objects

Combinatorics

184. The Contrapositive of '5n+2' is odd, then n is odd is

Answer: If n is even then 5n+2 is even

185. For All positive real numbers a and b, if a < b, then ____.

A^2 < b^2

186. The rational number '2' is the product of which of the two irrational numbers.

Under root (2), under root (2)

187. The sum of two integers is even when _____.

Both are even or both are odd

188. Proof by contraposition is based on the logical equivalence between a statement and its.

Contrapositive

189. The sum of two irrational numbers must be an irrational number.

False

190. In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

10

191. The sequence 1+ 2+ 3+ 4-----+ n is equal to _____.

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Answer: $n^2/2$

192. To prove by Mathematical induction basis step is_____.

Answer:

193. The____of a predicate variable is the set of all values that may be substituted in place of the variable.

Answer: Domain

194.

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